

M003 Extension 2 — English Worksheet (Advanced)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Week 3) · **Time:** about 45 minutes

Build **two mob heads** in 3D — a **creeper** and a **piglin**. Now every block needs **three** numbers: **(x, y, z)**. x is **across**, y is **up**, z is **forward** (depth). Both heads stand in front of you in the world — walk around them, then copy one block by block.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🕒

Read each set. Write what you will see.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. Does the line go across, up, or deeper?


```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Only y changes. Across, up, or deeper?


```
place green block at (3, 5, 1)
place black block at (3, 5, 1)
```

Same (x, y, z). Green, black, or both? Why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This block should sit one step **deeper**, at (3, 5, 2). It only has two numbers, so it lands on the front.

```
place green block at (3, 5)
```

Fixed code:

Why wrong? Why does your fix work?

Bug B — The eye should be **up high and one step deep**: (4, 7, 2). The last two numbers are swapped.

```
place black block at (4, 2, 7)
```

Fixed code:

Why wrong? Why does your fix work?

Bug C — Code is correct, but the whole head came out one step too far forward.

(the code matched the head)

What did your friend forget before pressing run?

3 · Build a Head

Pick **one** head in the world. Walk around it. Then build your own on your home spot.

Start with the front face (keep z the same, e.g. $z = 1$). Read each block's **x** (across) and **y** (up).

Then add depth. Build the same face again at $z = 2$, then $z = 3$, until your head is as deep as the real one.

💡 Tip: the eyes and mouth only show on the **front**. The sides and back are plain.

Send a photo. Then finish these — 4 to 6 sentences.

I built the ...

The lowest front block is at (x = ..., y = ..., z = ...).

To go up I changed the ... number; to go deeper I changed the ... number.

My head is ... blocks deep.

Once my head was wrong because ...

Next time I would ...

4 · Record Your Walkthrough 🎥

Film yourself showing your head. Use these words: **x, y, z, across, up, forward, home spot.**

1. Show your home spot. Why stand on it every time?
2. Walk around your head — show the front, a side, and the back.
3. Point at one block: "this one is at x ..., y ..., z ...".
4. Show one block you fixed.

Plan what you will say:

Submit ✓

Send this worksheet + your video on KakaoTalk.